

Def. Let  $V$  be a vector space over a field  $F$ .

A function  $f: V \times V \rightarrow F$  is called a bilinear form if: for all  $\alpha_1, \alpha_2, \beta_1, \beta_2 \in V$ ,  $c, d \in F$ ,

$$f(c \cdot \alpha_1 + d \cdot \alpha_2, \beta_1 + \beta_2) = c \cdot d \cdot f(\alpha_1, \beta_1) + c \cdot f(\alpha_1, \beta_2) + d \cdot f(\alpha_2, \beta_1) + f(\alpha_2, \beta_2).$$

Def. Let  $V$  be a fin. dim vector space over  $F$ , and let  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  be a basis for  $V$ .

Given a bilinear form  $f$ , define the matrix of  $f$  in  $\mathcal{B}$  as:

$$[f]_{\mathcal{B}} \stackrel{\text{def}}{=} (f(\alpha_i, \alpha_j))_{i,j}.$$

Thm. Let  $V$  be a finite-dimensional vector space over  $F$ , and  $\mathcal{B}$  be a basis.

Then, the space  $L(V, V, F) = \{f: V \times V \rightarrow F \mid f \text{ is bilinear form}\}$  is isomorphic to  $M_{nn}(F)$ .

Proof. Define  $\mathcal{Q}: L(V, V, F) \rightarrow M_{nn}(F)$ ;  $f \mapsto [f]_{\mathcal{B}}$ . Denote  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ .

Given  $\alpha = x_1 \alpha_1 + \dots + x_n \alpha_n$  and  $\beta = y_1 \alpha_1 + \dots + y_n \alpha_n$ ,

$$f(\alpha, \beta) = \sum_{i=1}^n x_i \cdot f(\alpha_i, \beta) = \sum_{i=1}^n x_i \cdot \sum_{j=1}^n y_j \cdot f(\alpha_i, \alpha_j) = \sum_{i=1}^n \sum_{j=1}^n x_i y_j \cdot f(\alpha_i, \alpha_j).$$

Thus, for  $X = [\alpha]_{\mathcal{B}}$  and  $Y = [\beta]_{\mathcal{B}}$ , one  $A = [f]_{\mathcal{B}}$ ,

$$f(\alpha, \beta) = X^t A Y = [\alpha]_{\mathcal{B}}^t [f]_{\mathcal{B}} [\beta]_{\mathcal{B}}.$$

Therefore, the bilinear form  $f$  is uniquely determined by the representation  $[f]_{\mathcal{B}}$ .

This proves the  $\mathcal{Q}$  is bijective. And, the  $\mathcal{Q}$  is linear, because

$$(c \cdot f + g)(\alpha_i, \alpha_j) = c \cdot f(\alpha_i, \alpha_j) + g(\alpha_i, \alpha_j), \text{ for any } c \in F \text{ and } f, g \in L(V, V, F).$$

Cor. If  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  is a basis for  $V$ , and  $\mathcal{B}^* = \{f_1, \dots, f_n\}$  is the dual basis, then  $f_{i,j}(\alpha, \beta) \stackrel{\text{def}}{=}} f_i(\alpha) \cdot f_j(\beta)$  ( $1 \leq i, j \leq n$ ) form a basis for  $V^*$ .